

# Costly Political Information and Equilibrium Coordination <sup>\*</sup>

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We revisit a standard model of costly information acquisition and aggregation in large majority elections. A large electorate has common preferences, but voters are uncertain which of two policies is better and may acquire costly private information before voting (Martinelli, 2006). Two insights emerge. First, information acquisition can exhibit strategic complementarities: when other voters acquire more information, a voter's own incentive to become informed may increase. We characterize when information acquisition is a strategic substitute and when it is a strategic complement. Second, these complementarities matter for efficiency: voters may coordinate on an efficient high-information equilibrium, but they may also coordinate on an inefficient low-information equilibrium. Our results suggest that ignorance in politics has a cultural component and is not purely driven by cost-benefit tradeoffs. We discuss implications for voter-information initiatives.

The Condorcet jury theorem and its modern versions (e.g. Austen-Smith and Banks, 1996; Feddersen and Pesendorfer, 1997) provide a central efficiency benchmark for democratic institutions: Even when relevant political information is dispersed among a large number of voters, majority elections lead to efficient outcomes under broad conditions. Piketty (1999) interprets this result as a political analog of the *First Welfare Theorem* for markets.

A challenge to the efficiency of political institutions is that political information is typically costly: In a large election, an individual voter has a negligible probability of affecting the outcome, and may therefore acquire too little information—or none at all—so efficiency can fail. This is the rational-ignorance problem emphasized by Downs (1957).

Martinelli (2006) made a fundamental observation about this problem, clarifying the conditions under which a version of the Condorcet jury theorem persists and

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“rational ignorance” does not preclude efficiency. Although each voter’s private incentive to acquire information becomes small in a large electorate, the election may still aggregate many small pieces of information. His analysis formalizes the horse race between the growing number of voters and the shrinking amount of information acquired by each individual voter. If the marginal cost of arbitrarily imprecise information becomes arbitrarily close to zero, there exists an equilibrium sequence in which the election outcome is asymptotically efficient. Subsequent work extends and refines this positive message in related environments (see, e.g., Martinelli, 2007; Triossi, 2013; Oliveros, 2013).

This paper revisits the setting of Martinelli (2006) and qualifies the usual rational-ignorance view. That view emphasizes the public-good nature of political information: because voters can free-ride on others’ information acquisition, political information is typically viewed as a strategic substitute.

Our first result shows that this view is incomplete. The acquisition of political information can also exhibit strategic complementarities: a local increase in the information acquired by other voters can raise a voter’s own incentive to become informed. We characterize the regions of the voters’ strategy space in which information acquisition is a strategic substitute and the regions in which it is a strategic complement (Proposition 1).

These complementarities have important efficiency consequences. Our second result shows that the efficient equilibrium sequences identified by Martinelli (2006) generically coexist with non-trivial inefficient equilibrium sequences in which information aggregation fails, even when arbitrarily imprecise information has arbitrarily low marginal cost; for example, this includes cost functions such as  $C(x) = e^{-\frac{1}{x}}$ . Voters may therefore coordinate on an inefficient low-information equilibrium. The costly-information version of the Condorcet jury theorem should consequently be read as a possibility result: efficiency obtains only under suitable equilibrium coordination. The condition for the inefficient sequence is generic: voters must not be exactly indifferent between the two alternatives under the prior (Theorem 1).

The coexistence of a low-information equilibrium and a high-information equilibrium is driven by the strategic complementarities of political information. In contrast, in the non-generic main specification of Martinelli (2006), equilibrium incentives are governed purely by a strategic-substitute logic; consequently, equilibria with information acquisition are unique, and coordination is not an issue.

Following Schelling (1980) and Myerson (2009), one may interpret equilibrium coordination as a reduced-form representation of political culture. Societies may

differ in the conventions, expectations, and focal points that make one equilibrium rather than another salient. In this sense, the low-information and high-information equilibria correspond to different cultures of political informedness.

Empirical work documents related differences in “cultures of news consumption.” News consumption varies not only with individual preferences and information costs, but also with contextual norms (Toff and Kalogeropoulos, 2020). For example, some citizens follow political news because they regard keeping informed as part of civic duty (McCombs and Poindexter, 1983; Palmer and Toff, 2020). By contrast, other citizens exhibit a “news-finds-me” orientation: they expect important political information to reach them through social media, peers, or everyday exposure without active search (Gil de Zúñiga *et al.*, 2017, 2020). These findings suggest that political informedness is shaped partly by shared norms about whether citizens should actively follow politics, and that social media may be associated with more passive information norms.

This perspective also offers a possible interpretation of the limited effects of standard voter-information campaigns, as documented in the meta-analysis by Dunning *et al.* (2019). Our model suggests an explanation: if information provision affects individual information costs but leaves citizens’ expectations about others’ information acquisition and surrounding norms largely unchanged, the electorate may remain coordinated on the low-information equilibrium. Voter-information initiatives may therefore need to do more than provide facts or lower individual information costs: they may also need to foster a culture of political informedness, in which active engagement with political information is expected.

**Related Literature.** This paper contributes to the literature on voting with costly information acquisition by identifying a strategic-complementarity channel. The usual rational-ignorance discussion emphasizes the strategic-substitute aspect of political information: voters can free-ride on the information acquired by others. We show that information acquisition can also exhibit strategic complementarities and that this creates an equilibrium-coordination problem with sharp efficiency implications. In other words, rational ignorance in politics has a cultural component.

In terms of the model, the paper is closest to the literature on the Condorcet jury theorem with endogenous information acquisition, especially the line of work following Martinelli (2006). Mukhopadhyaya (2003); Koriyama and Szentes (2009) also study related Condorcet environments and show that larger-than-optimal committee sizes may lead to social-welfare losses relative to smaller committees because

of free-riding, although these losses need not be large.

Other strands of the literature study related design questions (see, e.g., Persico, 2004; Gershkov and Szentes, 2009; Gerardi and Yariv, 2008). Because this literature typically focuses on efficient equilibria, our results motivate asking when additional non-trivial equilibria exist in these settings, possibly because of strategic complementarities in information acquisition. In such instances, questions of robust mechanism design naturally arise: For example, which voting mechanisms are worst-case optimal across all (non-trivial) equilibria?

An experimental literature has also evaluated testable implications of these theories; see, e.g., Grosser and Seebauer (2016); Elbittar *et al.* (2020); Mechtenberg and Tyran (2019); Bhattacharya *et al.* (2017). Our analysis invites follow-up work that tests the extent to which coordination problems matter empirically when voter information is endogenous.

The coexistence of a low-information equilibrium and a high-information equilibrium is reminiscent of the low-turnout and high-turnout equilibria in costly-voting models such as Palfrey and Rosenthal (1983). That literature shows that the high-turnout equilibria can be fragile to strategic uncertainty about preferences, costs, or the number of voters (Palfrey and Rosenthal, 1985; Myerson, 1998). A companion paper studies related information-acquisition games with strategic uncertainty about preferences, information costs, and prior beliefs, and shows that the same coordination problem persists (Heese, 2022). Thus, the multiplicities that arise under costly information and costly participation are substantially different.

# 1 Model

We restate the model of Martinelli (2006), largely following its original notation. There are  $2n + 1 \geq 3$  voters, two policies  $A$  and  $B$ , and a binary state  $z \in \{z_A, z_B\}$ . The voters hold a common prior. The prior probability of state  $z_A$  is  $q_A \in (0, 1)$ , and the prior probability of state  $z_B$  is  $q_B = 1 - q_A$ . Voters receive utility  $U(d, z)$  if the outcome is  $d$  and the state is  $z$ . We denote  $U(A, z_A) - U(B, z_A) = r_A$  and  $U(B, z_B) - U(A, z_B) = r_B$  and assume that  $r_A, r_B > 0$ . Thus, all voters prefer  $A$  in state  $z_A$  and  $B$  in  $z_B$ . We assume  $q_A r_A > q_B r_B$ , so that policy  $A$  maximizes expected utility under the prior.<sup>1</sup>

The timing is as follows. Each voter chooses the precision  $x \in [0, 1/2]$  of a binary private signal  $s \in \{s_A, s_B\}$ , where

$$\frac{1}{2} + x = \Pr(s_A \mid z_A) = \Pr(s_B \mid z_B).$$

Choosing precision  $x$  costs  $C(x)$ . The state and private signals then realize. After observing their private signals, all voters vote simultaneously, either for  $A$  or for  $B$ . Finally, the outcome is decided by simple majority rule.

We assume that  $C$  is strictly increasing, strictly convex, twice continuously differentiable on  $(0, \frac{1}{2})$  and that  $C(0) = 0$ ; in other words, acquiring no information is costless. Note that  $C''(0) \in [0, \infty)$  exists. Whenever  $\lim_{x \downarrow 0} C'''(x)$  exists, we write  $C'''(0) := \lim_{x \downarrow 0} C'''(x)$ , allowing  $C'''(0) = \infty$  when this limit diverges.

A pure strategy is a triple  $(x, v_A, v_B)$ , where  $x \in [0, 1/2]$  specifies the chosen precision,  $v_A$  specifies which policy the voter votes for after observing signal  $s_A$ , and  $v_B$  specifies which policy the voter votes for after observing signal  $s_B$ . A mixed strategy  $\alpha$  is a probability distribution over pure strategies. We consider symmetric mixed-strategy Bayes–Nash equilibria. An equilibrium in which some strategy  $(x, v_A, v_B)$  with  $x > 0$  is in the support is called an *equilibrium with information acquisition*. When no confusion arises, we simply write *equilibrium*.

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<sup>1</sup>The case  $q_A r_A < q_B r_B$  yields qualitatively the same results with the roles of  $A$  and  $B$  reversed. In the knife-edge case  $q_A r_A = q_B r_B$ , the low-information equilibrium sequence identified in the main result (Theorem 1) does not exist.

## 2 Equilibrium Representation

We provide a compact characterization of equilibria with information acquisition in large electorates. To state the result, we first explain the “pivotal calculus” of strategic voters.

Fix a voter and the strategy  $\alpha$  of all other voters. Whenever the votes of the other voters do not split into  $n$  votes for  $A$  and  $n$  votes for  $B$ , the fixed voter’s choice of her strategy does not affect the outcome. Thus, when comparing expected utilities from two strategies, it suffices to consider the pivotal event in which the votes of the other voters split into  $n$  votes for  $A$  and  $n$  votes for  $B$ . Denote by  $\pi(z; \alpha)$  the probability that a voter votes for  $A$  in state  $z \in \{z_A, z_B\}$  under  $\alpha$ . The likelihood of the pivotal event is

$$\Pr(\text{piv} \mid z; \alpha) = \binom{2n}{n} \left( \pi(z; \alpha)(1 - \pi(z; \alpha)) \right)^n. \quad (1)$$

Our result characterizes equilibria as strategies  $\alpha(x, \delta)$  that mix between  $(x, A, B)$  with probability  $1 - \delta$  and  $(0, A, A)$  with probability  $\delta > 0$  and satisfy two relevant equilibrium conditions.

**Lemma 1** *Assume  $q_A r_A > q_B r_B$  and  $C'(0) = 0$ . For all sufficiently large  $n$ , a strategy  $\alpha$  is an equilibrium with information acquisition if and only if there is a pair  $(x, \delta) \in (0, \frac{1}{2}) \times (0, 1)$  so that  $\alpha = \alpha(x, \delta)$  and  $\alpha$  satisfies*

$$\left( \Pr(\text{piv} \mid z_A; \alpha) q_A r_A + \Pr(\text{piv} \mid z_B; \alpha) q_B r_B \right) \left( \frac{1}{2} + x \right) - C(x) = \Pr(\text{piv} \mid z_A; \alpha) q_A r_A, \quad (2)$$

$$\Pr(\text{piv} \mid z_A; \alpha) q_A r_A + \Pr(\text{piv} \mid z_B; \alpha) q_B r_B = C'(x). \quad (3)$$

The formal proof builds on arguments already present in Martinelli (2006) and appears in the appendix. The first equilibrium condition states that voters are indifferent between  $(x, A, B)$  and  $(0, A, A)$ . To see this, compare the expected utilities from the two strategies. Up to the constant utility from non-pivotal events, the expected utility from choosing  $(x, A, B)$  is

$$\begin{aligned} & \Pr(\text{piv} \mid z_A; \alpha) q_A \left( U(A, z_A) \left( \frac{1}{2} + x \right) + U(B, z_A) \left( \frac{1}{2} - x \right) \right) \\ & + \Pr(\text{piv} \mid z_B; \alpha) q_B \left( U(B, z_B) \left( \frac{1}{2} + x \right) + U(A, z_B) \left( \frac{1}{2} - x \right) \right) - C(x). \end{aligned}$$

The expected utility from  $(0, A, A)$  is, up to the same constant,

$$\Pr(\text{piv} \mid z_A; \alpha)q_A U(A, z_A) + \Pr(\text{piv} \mid z_B; \alpha)q_B U(A, z_B).$$

Subtracting  $\Pr(\text{piv} \mid z_A; \alpha)q_A U(B, z_A) + \Pr(\text{piv} \mid z_B; \alpha)q_B U(A, z_B)$  from both expressions yields the stated indifference condition.

The second equilibrium condition states that the marginal cost of acquiring precision  $x > 0$  equals the marginal benefit. Equivalently, it is the first-order condition for the optimal precision.

Under the mixed strategy  $\alpha(x, \delta)$ , a voter chooses  $(x, A, B)$  with probability  $1 - \delta$  and  $(0, A, A)$  with probability  $\delta$ . The induced voting probabilities are

$$\begin{aligned}\pi(z_A; \alpha(x, \delta)) &= (1 - \delta) \left( \frac{1}{2} + x \right) + \delta, \\ \pi(z_B; \alpha(x, \delta)) &= (1 - \delta) \left( \frac{1}{2} - x \right) + \delta.\end{aligned}\tag{4}$$

The prior bias toward  $A$ , captured by  $q_A r_A > q_B r_B$ , shapes the qualitative form of equilibria. It implies that all equilibria are shifted toward voting for  $A$  in the sense that all uninformed voters choose  $A$ . A prior bias toward  $B$ , i.e.  $q_A r_A < q_B r_B$ , would imply that, in all equilibria with information acquisition, the uninformed voters choose  $B$ . Thus, these equilibria involve mixing between  $(x, A, B)$  and  $(0, B, B)$  instead of mixing between  $(x, A, B)$  and  $(0, A, A)$ .

In what follows, we identify equilibria with information acquisition by their representing pairs  $(x, \delta)$ .

### 3 Complements and Substitutes

The usual rational-ignorance argument emphasizes the strategic-substitute aspect of political information: voters underinvest because they can free-ride on the information acquired by others. We qualify this argument by showing that the incentives for information acquisition can contain both strategic-substitute and strategic-complement effects.

We say that information acquisition is locally a *strategic complement* (*substitute*) at  $\alpha(x, \delta)$  if the marginal benefit of information is locally increasing (respectively, decreasing) in the precision  $x$  chosen by other voters:

$$\frac{\partial \text{MB}_n(x, \delta)}{\partial x} > 0 \quad (\text{respectively, } < 0),$$

$$\text{for } \text{MB}_n(x, \delta) = \Pr(\text{piv} \mid z_A; \alpha(x, \delta))q_{Ar_A} + \Pr(\text{piv} \mid z_B; \alpha(x, \delta))q_{Br_B}.$$

The next result characterizes when information acquisition is locally a strategic complement and when it is locally a strategic substitute.

**Proposition 1** *Suppose  $C'(0) = 0$  and  $q_{Ar_A} > q_{Br_B}$ , and consider the voters' strategy  $\alpha(x, \delta)$ . For every sufficiently large  $n$ , there exists a threshold  $\delta_{1,n} \in (0, 1/2)$  such that*

1. *for  $0 \leq \delta < \delta_{1,n}$ , information acquisition is locally a strategic substitute at  $\alpha(x, \delta)$  for all  $x \in (0, \frac{1}{2})$ ;*
2. *for  $\delta_{1,n} < \delta < 1/2$ , information acquisition is locally a strategic substitute, then a strategic complement, and then again a strategic substitute, as  $x$  varies from 0 to  $\frac{1}{2}$ ;*
3. *for  $1/2 < \delta < 1$ , information acquisition is locally a strategic substitute and then a strategic complement, as  $x$  varies from 0 to  $\frac{1}{2}$ .*

The proof is in the appendix. The marginal benefit of information is best understood through its connection to the *margin of victory* in the two states,

$$\begin{aligned} \text{MV}(z; x, \delta) &= \left| \pi(z; \alpha(x, \delta)) - \frac{1}{2} \right| \\ &= \begin{cases} \delta + (1 - \delta) \left( \frac{1}{2} + x \right) - \frac{1}{2} & \text{for } z = z_A, \\ \left| \delta + (1 - \delta) \left( \frac{1}{2} - x \right) - \frac{1}{2} \right| & \text{for } z = z_B. \end{cases} \end{aligned}$$



The marginal benefit increases in the pivotal likelihoods  $\Pr(\text{piv} \mid z; \alpha(x, \delta))$ , which are themselves monotone decreasing in the margin of victory, as

$$\Pr(\text{piv} \mid z; \alpha(x, \delta)) = \binom{2n}{n} \left( \frac{1}{4} - \text{MV}(z; x, \delta)^2 \right)^n. \quad (5)$$

Overall, a larger margin of victory implies a smaller marginal benefit of information.

We now explain the three regimes using this connection to the margins of victory.

When  $\delta = 0$ , the margin of victory increases in  $x$  in both states. Thus, the value of information is monotonically decreasing, corresponding to substitutes. This monotonicity extends to values of  $\delta$  in a whole interval  $[0, \delta_{1,n})$ , by continuity.

When  $\delta_{1,n} < \delta < 1$ , the vote share in  $z_A$  is larger than  $\frac{1}{2}$  for all  $x$  but the vote share in  $z_B$  is larger than  $\frac{1}{2}$  only if

$$x < x'' \text{ for } x'' = \frac{\delta}{2(1 - \delta)}.$$

For  $x < x''$ , an increase in  $x$  has two countervailing effects: it increases the margin of victory in  $z_A$ , but it decreases it in  $z_B$ . The proof shows that the first effect dominates on an interval  $[0, x']$ , where  $0 < x' < \min\{\frac{1}{2}, x''\}$ , and the second effect dominates on  $[x', \min\{\frac{1}{2}, x''\}]$ . If  $\delta > \frac{1}{2}$ , then  $x'' > \frac{1}{2}$  and these two intervals imply the overall substitutes-complements pattern. If  $\delta < \frac{1}{2}$ , there is another interval  $[x'', \frac{1}{2}]$  on which the margin of victory increases with  $x$  in both states, implying an overall substitutes-complements-substitutes pattern.

Figure 1 below illustrates the three regimes in an example.

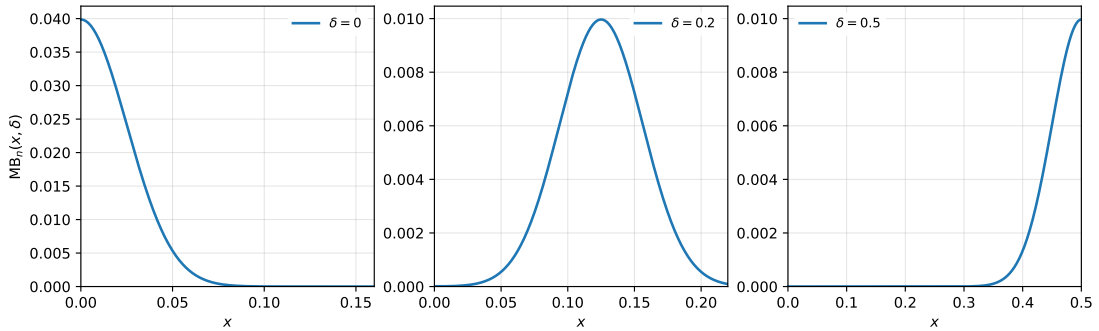


Figure 1: The marginal benefit  $\text{MB}_n(x, \delta)$  of information as a function of  $x \in (0, \frac{1}{2})$  for fixed  $\delta = 0$  (left),  $\delta = 0.2$  (middle), and  $\delta = 0.5$  (right), in an example with  $n = 200$ ,  $r_A = r_B = 1$ ,  $q_A = 3/4$ , and  $q_B = 1/4$ .

## 4 The Voters' Coordination Problem

We characterize the outcomes of large elections ( $n \rightarrow \infty$ ). An equilibrium sequence  $(\alpha_n)_{n \in \mathbb{N}}$  has *asymptotically efficient outcomes* if the probability that  $A$  is elected in state  $z_A$  and the probability that  $B$  is elected in  $z_B$  both converge to one as  $n \rightarrow \infty$ . It has *asymptotically inefficient outcomes* otherwise.

We focus on situations in which efficiency is possible. As shown by Martinelli (2006), a sequence of equilibria with asymptotically efficient outcomes exists if and only if  $C'(0) = C''(0) = 0$ . This condition means that the marginal cost of arbitrarily imprecise information is arbitrarily low. In the efficient sequence, individual information acquisition becomes arbitrarily imprecise, but slowly enough for the aggregate information—acquired by an increasing number of voters—to enable asymptotically efficient outcomes. Under the same condition, aggregate information costs converge to zero. Thus, the equilibrium sequence maximizes utilitarian welfare even after accounting for information costs (cf. Theorem 3 and Theorem 4 in Martinelli (2006)).

We show that, whenever efficiency is possible, the voters face a coordination problem. They may miscoordinate on an equilibrium sequence with asymptotically inefficient outcomes. In the inefficient equilibria, individual information acquisition is so low that information aggregation fails; that is, an outside observer cannot learn the state from observing the realized vote share, as  $n \rightarrow \infty$ .

The next result characterizes the possible limiting outcomes of equilibrium sequences.

**Theorem 1** *Assume  $C'(0) = C''(0) = 0$ .*

1. *There exists a sequence of equilibria with asymptotically efficient outcomes.*
2. *There exists a sequence of equilibria with information acquisition that has asymptotically inefficient outcomes. Along this sequence, the prior-favored policy  $A$  is elected with probability converging to one in both states.*
3. *Any equilibrium sequence with information acquisition and convergent outcome probabilities has one of two limit outcomes: either it is asymptotically efficient, or the prior-favored policy  $A$  is elected with probability converging to one in both states.*

The proof of Theorem 1 is in the appendix. It constructs the equilibrium sequences using a fixed-point theorem, established in the companion paper Ekmekci

*et al.* (2025), that extends the Poincaré–Miranda theorem.<sup>2</sup> Economically, the co-existence of the low-information and high-information equilibria is driven by the strategic complementarities in information acquisition described in Proposition 1.

To get some intuition, it is instructive to compare the “asymmetric” setting considered here, in which  $A$  is the prior-favored policy, with the “symmetric” main setting in Martinelli (2006) in which the voters are exactly indifferent given the prior. In the symmetric setting, equilibrium incentives are governed by a pure strategic-substitute logic and no coordination problem arises, unlike in the asymmetric setting.

To explain this, we describe equilibria in both settings through a one-dimensional fixed-point equation and then illustrate the role of complementarities in an example. In the asymmetric setting, for each  $x \in (0, \frac{1}{2})$  such that the indifference condition (2) has a solution, let  $\delta(x)$  denote the minimal such solution. Denote by  $\hat{x}_n(x, \delta(x))$  the precision solving the first-order condition (3) given  $\alpha(x, \delta(x))$ . Any fixed point

$$x = \hat{x}_n(x, \delta(x)) \tag{6}$$

of the correspondence  $x \mapsto \hat{x}_n(x, \delta(x))$  induces a strategy  $\alpha(x, \delta(x))$  satisfying both equilibrium conditions (2) and (3), hence an equilibrium. In the symmetric setting, the symmetry implies that all equilibria satisfy  $\delta = 0$  and are characterized by the first-order condition (3), which we rewrite as

$$x = \hat{x}_n(x, 0); \tag{7}$$

cf. Theorem 1 in Martinelli (2006).

Figure 2 illustrates the two fixed-point equations (6) and (7) in an example. The right panel shows the mapping  $x \mapsto \hat{x}_n(x, 0)$  in the symmetric setting and the left panel shows the mapping  $x \mapsto \hat{x}_n(x, \delta(x))$  in the asymmetric setting.

In the symmetric setting,  $\delta = 0$  implies that information acquisition is locally a strategic substitute (cf. Proposition 1). As a consequence, the precision  $\hat{x}_n(x, 0)$  of the best response to  $\alpha(x, 0)$  is strictly decreasing in  $x$ : for a given voter, the more information the other voters acquire, the less incentive the voter has to acquire information herself. The monotonicity implies a unique crossing point with the identity map and thus a unique equilibrium. Absent complementarities, equilibrium is unique.

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<sup>2</sup>The original Poincaré–Miranda theorem appears in Miranda (1940).

In the asymmetric setting, the complementarities imply a non-monotone shape of the fixed-point mapping  $x \mapsto \hat{x}_n(x, \delta(x))$ . This enables two crossing points with the identity map, and thus two equilibria. At the high-information equilibrium, voters acquire more information and, relative to the low-information equilibrium, but also have stronger incentives to acquire information under the best response.

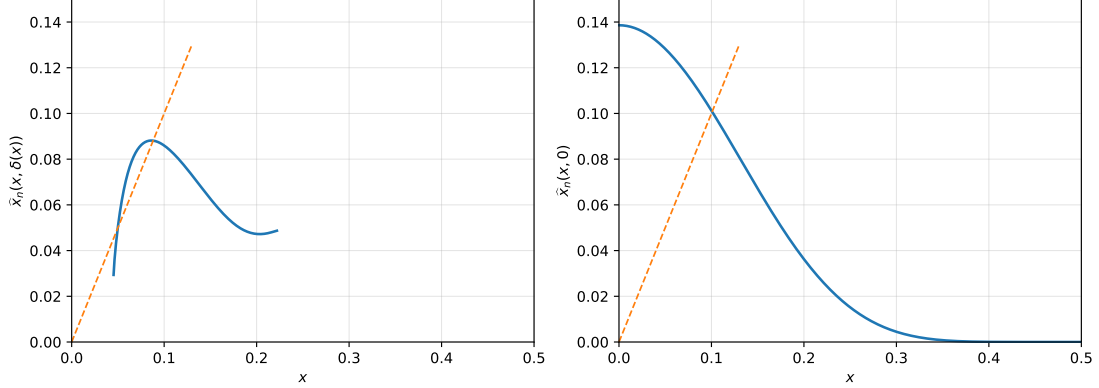


Figure 2: Example with  $n = 10$ ,  $C(x) = x^{2.3}$ , and  $r_A = r_B = 1$ . The left panel shows the fixed-point mapping  $x \mapsto \hat{x}_n(x, \delta(x))$  for the asymmetric setting with  $q_A = 3/4$ ; the right panel shows the fixed-point mapping  $x \mapsto \hat{x}_n(x, 0)$  for the symmetric setting in which  $q_A = 1/2$ .

The asymptotic inefficiency of the low-information equilibrium can be seen through the equilibrium indifference condition (2) that ensures the voters have a best response with information acquisition. As  $x \rightarrow 0$  in all equilibrium sequences, indifference between  $(x, A, B)$  and  $(0, A, A)$  requires asymptotic indifference between  $(0, A, B)$  and  $(0, A, A)$  and thus between  $(0, A, A)$  and  $(0, B, B)$ ; hence,

$$\frac{q_A r_A}{q_B r_B} \frac{\Pr(\text{piv} \mid z_A; \alpha_n)}{\Pr(\text{piv} \mid z_B; \alpha_n)} \rightarrow 1.$$

Because  $q_A r_A > q_B r_B$ , the pivotal event must be asymptotically more likely in state  $z_B$  than in state  $z_A$ . Using (5), we can rewrite the condition as

$$\frac{q_B r_B}{q_A r_A} = \lim_{n \rightarrow \infty} \left( \frac{\frac{1}{4} - \text{MV}(z_A; x_n, \delta_n)^2}{\frac{1}{4} - \text{MV}(z_B; x_n, \delta_n)^2} \right)^n.$$

Thus the indifference condition requires the election to be closer in state  $z_B$  than in state  $z_A$ .

However, this does not require  $B$  to have an expected majority in  $z_B$ . Under

$\alpha(x, \delta)$ ,

$$\pi(z_A; \alpha(x, \delta)) - \frac{1}{2} = \frac{\delta}{2} + (1 - \delta)x, \quad \pi(z_B; \alpha(x, \delta)) - \frac{1}{2} = \frac{\delta}{2} - (1 - \delta)x.$$

The high-information equilibrium has

$$\frac{\delta}{2} - (1 - \delta)x < 0,$$

so  $B$  has an expected majority in state  $z_B$ . The low-information equilibrium instead has

$$0 < \frac{\delta}{2} - (1 - \delta)x < \frac{\delta}{2} + (1 - \delta)x.$$

In this case, the pivotal event is still more likely in  $z_B$  than in  $z_A$ , because the election is closer in  $z_B$ . This supports the indifference condition and information acquisition despite the prior bias toward  $A$ . But the expected majority is still for  $A$  in both states, preventing  $B$  from being elected with probability converging to one and therefore precluding asymptotically efficient outcomes.

**Remarks.** Two remarks are worth noting. First, the coordination problem is present even in small electorates. The example in Figure 2 demonstrates existence for an electorate of just 21 voters. Second, the inefficient equilibrium sequence exists even when the information costs vanish arbitrarily fast to zero as  $x \rightarrow 0$ . Information costs may even vanish exponentially fast; for example, the result applies to  $C(x) = \exp(-1/x)$  for  $x \in (0, \frac{1}{2}]$  and  $C(0) = 0$ .

## A Basic Notation

Fix a symmetric strategy  $\alpha$  used by the other  $2n$  voters and write

$$\rho_A = q_A r_A, \quad \rho_B = q_B r_B,$$

so that  $\rho_A > \rho_B$ , and

$$P_A = \Pr(\text{piv} \mid z_A; \alpha), \quad P_B = \Pr(\text{piv} \mid z_B; \alpha),$$

and

$$R_A = P_A q_A r_A, \quad R_B = P_B q_B r_B.$$

For a pair  $(x, \delta)$ , define

$$a_A(x, \delta) = \frac{\delta}{2} + (1 - \delta)x, \quad a_B(x, \delta) = \frac{\delta}{2} - (1 - \delta)x. \quad (8)$$

Then

$$\pi(z_A; \alpha(x, \delta)) = \frac{1}{2} + a_A(x, \delta), \quad \pi(z_B; \alpha(x, \delta)) = \frac{1}{2} + a_B(x, \delta).$$

## B Proof of Lemma 1

We prove the lemma in three steps. First, we show that, for all sufficiently large  $n$ , every equilibrium with information acquisition has support

$$\{(x, A, B), (0, A, A)\}$$

for some  $x \in (0, \frac{1}{2})$ , so it is a strategy  $\alpha(x, \delta)$  for some  $(x, \delta) \in (0, \frac{1}{2}) \times (0, 1)$ . Second, we show that every such equilibrium satisfies the two conditions in Lemma 1, namely (2) and (3). Third, we prove the converse: if  $(x, \delta) \in (0, \frac{1}{2}) \times (0, 1)$  is such that  $\alpha(x, \delta)$  satisfies (2) and (3), then  $\alpha(x, \delta)$  is an equilibrium with information acquisition.

**Step 1: The support of equilibria with information acquisition.** Fix a voter and a symmetric strategy  $\alpha$  used by the other  $2n$  voters. Only pivotal events matter for payoff comparisons. Up to terms that are constant across the fixed

voter's strategies, the payoff from choosing  $(x_i, A, B)$  is

$$V_I(x_i; \alpha) = (R_A + R_B) \left( \frac{1}{2} + x_i \right) - C(x_i).$$

The payoffs from the two uninformative pure-voting strategies are

$$V_A(\alpha) = R_A, \quad V_B(\alpha) = R_B.$$

For any  $x_i > 0$ , the strategies  $(x_i, A, A)$  and  $(x_i, B, B)$  are strictly dominated by  $(0, A, A)$  and  $(0, B, B)$ , respectively, because they induce the same vote but add a positive information cost. The strategy  $(x_i, B, A)$  is strictly dominated by  $(x_i, A, B)$  because the signal is informative and voting with the signal raises the probability of choosing the correct policy in each state. Hence any pure strategy with information acquisition that is used with positive probability in equilibrium must be of the form  $(x_i, A, B)$ .

The function  $V_I(\cdot; \alpha)$  is strictly concave because  $C$  is strictly convex. Moreover, in any best response with information acquisition,  $R_A + R_B > 0$ , and therefore

$$V'_I(0; \alpha) = R_A + R_B - C'(0) > 0.$$

Thus the best response uses at most one informative precision. For large enough  $n$ , this precision is interior. Indeed, for any fixed  $\eta \in (0, 1/2)$ , the pivotal likelihood is uniformly bounded above by

$$\binom{2n}{n} 4^{-n},$$

which converges to zero. Hence, uniformly over  $\alpha$ ,

$$R_A + R_B \leq (q_A r_A + q_B r_B) \binom{2n}{n} 4^{-n} < C'(\eta)$$

for all sufficiently large  $n$ . Strict concavity then rules out any maximizer  $x_i \geq \eta$ . Since  $V'_I(0; \alpha) > 0$ , the informative maximizer is positive. Thus, for all sufficiently large  $n$ , the optimal informative precision  $x^*(\alpha)$  lies in  $(0, 1/2)$  and satisfies

$$R_A + R_B = C'(x^*(\alpha)).$$

Since the payoff from  $(0, A, B)$  is a convex combination of  $V_A(\alpha)$  and  $V_B(\alpha)$ , and the same is true of  $(0, B, A)$ , neither signal-contingent strategy with zero precision can be strictly better than both uninformative pure-voting strategies. If

$V_A(\alpha) = V_B(\alpha)$ , then  $V_I(0; \alpha) = V_A(\alpha) = V_B(\alpha)$  and  $V_I'(0; \alpha) > 0$ , so the maximized informative payoff is strictly higher than both uninformative payoffs. If  $V_A(\alpha) \neq V_B(\alpha)$ , only the better uninformative pure-voting strategy can possibly be tied with the informative strategy. Hence a best response with information acquisition can have only one of the following supports:

$$\{(x^*, A, B)\}, \quad \{(x^*, A, B), (0, A, A)\}, \quad \{(x^*, A, B), (0, B, B)\}.$$

We now rule out the first and third possibilities for all sufficiently large  $n$  when  $q_A r_A > q_B r_B$ .

First consider the candidate equilibrium in which all voters use  $(x, A, B)$ . Then  $P_A = P_B = P$  and the interior first-order condition is

$$P(q_A r_A + q_B r_B) = C'(x).$$

The informative strategy must also be at least as good as always voting for  $A$ :

$$P(q_A r_A + q_B r_B) \left( \frac{1}{2} + x \right) - C(x) \geq P q_A r_A.$$

Using the first-order condition to substitute for  $P(q_A r_A + q_B r_B)$  gives

$$C'(x) \left( \frac{1}{2} + x \right) - C(x) \geq C'(x) \frac{q_A r_A}{q_A r_A + q_B r_B}.$$

Since  $x > 0$  and  $C'(x) > 0$ , division by  $C'(x)$  yields

$$\frac{1}{2} + x - \frac{C(x)}{C'(x)} \geq \frac{q_A r_A}{q_A r_A + q_B r_B}.$$

Subtracting  $\frac{1}{2}$  from both sides gives

$$x - \frac{C(x)}{C'(x)} \geq \frac{q_A r_A}{q_A r_A + q_B r_B} - \frac{1}{2}.$$

The right-hand side simplifies as

$$\frac{q_A r_A}{q_A r_A + q_B r_B} - \frac{1}{2} = \frac{2q_A r_A - (q_A r_A + q_B r_B)}{2(q_A r_A + q_B r_B)} = \frac{1}{2} \frac{q_A r_A - q_B r_B}{q_A r_A + q_B r_B}.$$



Rearranging thus gives

$$x - \frac{1}{2} \frac{q_A r_A - q_B r_B}{q_A r_A + q_B r_B} \geq \frac{C(x)}{C'(x)} \geq 0.$$

Thus  $x$  must be bounded away from zero. But the first-order condition and the uniform convergence of pivotal probabilities to zero imply  $C'(x) \rightarrow 0$ , hence  $x \rightarrow 0$ , along any sequence of pure informative equilibria. This contradiction rules out the pure informative case for all sufficiently large  $n$ .

Next consider a candidate equilibrium with support  $\{(x, A, B), (0, B, B)\}$ . Let  $\beta \in (0, 1)$  be the probability assigned to  $(0, B, B)$ . Then

$$\pi(z_A; \alpha) = (1 - \beta) \left( \frac{1}{2} + x \right), \quad \pi(z_B; \alpha) = (1 - \beta) \left( \frac{1}{2} - x \right).$$

Set

$$a = (1 - \beta) \left( \frac{1}{2} + x \right), \quad b = (1 - \beta) \left( \frac{1}{2} - x \right).$$

Then  $\pi(z_A; \alpha) = a$  and  $\pi(z_B; \alpha) = b$ , so

$$\begin{aligned} \pi(z_A; \alpha)(1 - \pi(z_A; \alpha)) - \pi(z_B; \alpha)(1 - \pi(z_B; \alpha)) &= a(1 - a) - b(1 - b) \\ &= (a - b)(1 - a - b). \end{aligned}$$

Since

$$a - b = 2(1 - \beta)x \quad \text{and} \quad 1 - a - b = 1 - (1 - \beta) = \beta,$$

we obtain

$$\pi(z_A; \alpha)(1 - \pi(z_A; \alpha)) - \pi(z_B; \alpha)(1 - \pi(z_B; \alpha)) = 2\beta(1 - \beta)x > 0.$$

Hence  $P_A > P_B$ , given (1). Together with  $q_A r_A > q_B r_B$ , this gives

$$V_A(\alpha) = P_A q_A r_A > P_B q_B r_B = V_B(\alpha).$$

Thus  $(0, B, B)$  cannot be in the support of a best response. By way of exclusion, the support of any equilibrium with information acquisition is therefore

$$\{(x, A, B), (0, A, A)\}.$$

**Step 2: Necessity of the two equilibrium conditions.** Consider any equilibrium with information acquisition. By the first step, when  $n$  is sufficiently

large, the equilibrium is a strategy  $\alpha(x, \delta)$  with  $(x, \delta) \in (0, \frac{1}{2}) \times (0, 1)$  and support  $\{(x, A, B), (0, A, A)\}$ . Since  $x$  is interior and the strategy  $(x, A, B)$  is a best response, it must satisfy the first-order condition

$$\Pr(\text{piv} \mid z_A; \alpha(x, \delta))q_A r_A + \Pr(\text{piv} \mid z_B; \alpha(x, \delta))q_B r_B = C'(x),$$

which is precisely (3). Since both  $(x, A, B)$  and  $(0, A, A)$  are used with positive probability, they must yield the same payoff. The indifference condition is precisely (2).

**Step 3: Sufficiency.** It remains to prove the converse. Suppose that  $(x, \delta) \in (0, 1/2) \times (0, 1)$  is so that the strategy  $\alpha(x, \delta)$  satisfies (2) and (3). The strategy  $\alpha(x, \delta)$  has support  $\{(x, A, B), (0, A, A)\}$ , and (2) states that these two strategies yield the same payoffs. In the following, we argue that all strategies outside the support have weakly lower payoffs.

First, we show that  $(0, B, B)$  is strictly worse. By (3),

$$R_A + R_B = C'(x).$$

By the indifference condition (2),

$$R_A = (R_A + R_B) \left( \frac{1}{2} + x \right) - C(x).$$

Substituting  $R_A + R_B = C'(x)$  gives

$$R_A = C'(x) \left( \frac{1}{2} + x \right) - C(x).$$

Therefore,

$$R_B = C'(x) - R_A = C'(x) - C'(x) \left( \frac{1}{2} + x \right) + C(x) = C'(x) \left( \frac{1}{2} - x \right) + C(x).$$

Hence

$$\begin{aligned} V_A(\alpha(x, \delta)) - V_B(\alpha(x, \delta)) &= R_A - R_B \\ &= \left[ C'(x) \left( \frac{1}{2} + x \right) - C(x) \right] - \left[ C'(x) \left( \frac{1}{2} - x \right) + C(x) \right] \\ &= 2(xC'(x) - C(x)) > 0. \end{aligned}$$

The strict inequality follows from strict convexity of  $C$  and  $C(0) = 0$ : for  $x > 0$ ,

$$C(x) = \int_0^x C'(t) dt < xC'(x).$$

Thus  $(0, B, B)$  is strictly worse than the two strategies in the support.

Second, the uninformative signal-contingent strategies  $(0, A, B)$  and  $(0, B, A)$  are strictly worse because their payoffs are convex combinations of the payoffs  $R_A$  and  $R_B$  from  $(0, A, A)$  and  $(0, B, B)$ ; they are given by

$$\frac{1}{2}(R_A + R_B).$$

Third, as argued above, any costly strategy other than voting with the signal is dominated. Fourth, the payoff  $V_I(y; \alpha(x, \delta))$  of any strategy  $(y, A, B)$  with  $y \neq x$  yields a weakly lower payoff than  $(x, A, B)$ , because under  $\alpha(x, \delta)$ , the payoff  $V_I(\cdot; \alpha(x, \delta))$  is strictly concave, and (3) makes  $x$  its unique maximizer.

We conclude that every strategy in the support of  $\alpha(x, \delta)$  is a best response. Hence  $\alpha(x, \delta)$  is a symmetric equilibrium with information acquisition.

## C Proof of Proposition 1

For a strategy  $\alpha(x, \delta)$ , the marginal benefit of a higher precision is

$$\text{MB}_n(x, \delta) = \binom{2n}{n} \left[ \rho_A \left( \frac{1}{4} - a_A(x, \delta)^2 \right)^n + \rho_B \left( \frac{1}{4} - a_B(x, \delta)^2 \right)^n \right].$$

Since

$$\frac{\partial a_A(x, \delta)}{\partial x} = 1 - \delta, \quad \frac{\partial a_B(x, \delta)}{\partial x} = -(1 - \delta),$$

differentiating  $\text{MB}_n$  with respect to  $x$  gives

$$\begin{aligned} \frac{\partial \text{MB}_n(x, \delta)}{\partial x} = 2n(1 - \delta) \binom{2n}{n} & \left[ \rho_B a_B(x, \delta) \left( \frac{1}{4} - a_B(x, \delta)^2 \right)^{n-1} \right. \\ & \left. - \rho_A a_A(x, \delta) \left( \frac{1}{4} - a_A(x, \delta)^2 \right)^{n-1} \right]. \end{aligned}$$

The factor outside the square brackets is strictly positive. Hence the sign of  $\partial \text{MB}_n(x, \delta) / \partial x$  is the sign of the expression inside the square brackets.

If  $\delta = 0$ , then  $a_A(x, 0) = x$  and  $a_B(x, 0) = -x$ . The expression inside the square

brackets becomes

$$-(\rho_A + \rho_B)x \left( \frac{1}{4} - x^2 \right)^{n-1} < 0$$

for every  $x \in (0, 1/2)$ . Thus  $\text{MB}_n(\cdot, 0)$  is strictly decreasing. In the remainder of the proof, fix  $\delta > 0$ .

**Claim 1** *If  $a_B(x, \delta) \leq 0$ , then*

$$\frac{\partial \text{MB}_n(x, \delta)}{\partial x} < 0.$$

*If  $a_B(x, \delta) > 0$  and*

$$s = \frac{2(1 - \delta)x}{\delta},$$

*then the sign of  $\partial \text{MB}_n(x, \delta)/\partial x$  is the sign of  $\Lambda_n(s, \delta)$ , where*

$$\Lambda_n(s, \delta) = \log \frac{\rho_B}{\rho_A} + \log \frac{1 - s}{1 + s} + (n - 1) \log \frac{1 - \delta^2(1 - s)^2}{1 - \delta^2(1 + s)^2}.$$

**Proof.** If  $a_B(x, \delta) \leq 0$ , then the first term inside the square brackets in the derivative of  $\text{MB}_n$  is weakly negative, while the second term is strictly negative since  $a_A(x, \delta) > 0$ . Hence the overall derivative is strictly negative.

Now suppose  $a_B(x, \delta) > 0$ . Then  $s < 1$ , and

$$a_A(x, \delta) = \frac{\delta}{2}(1 + s), \quad a_B(x, \delta) = \frac{\delta}{2}(1 - s).$$

The derivative  $\partial_x \text{MB}_n(x, \delta)$  is positive if and only if

$$\rho_B a_B(x, \delta) \left( \frac{1}{4} - a_B(x, \delta)^2 \right)^{n-1} > \rho_A a_A(x, \delta) \left( \frac{1}{4} - a_A(x, \delta)^2 \right)^{n-1}.$$

Taking logs, the inequality is equivalent to

$$\log \frac{\rho_B}{\rho_A} + \log \frac{a_B(x, \delta)}{a_A(x, \delta)} + (n - 1) \log \frac{\frac{1}{4} - a_B(x, \delta)^2}{\frac{1}{4} - a_A(x, \delta)^2} > 0.$$

( $x < 1/2$  implies  $a_A(x, \delta) < 1/2$ , so all logarithms below are well defined.) Using

$$\frac{a_B(x, \delta)}{a_A(x, \delta)} = \frac{1 - s}{1 + s},$$

and

$$\frac{\frac{1}{4} - a_B(x, \delta)^2}{\frac{1}{4} - a_A(x, \delta)^2} = \frac{\frac{1}{4} - \frac{\delta^2}{4}(1-s)^2}{\frac{1}{4} - \frac{\delta^2}{4}(1+s)^2} = \frac{1 - \delta^2(1-s)^2}{1 - \delta^2(1+s)^2},$$

the preceding inequality becomes exactly  $\Lambda_n(s, \delta) > 0$ . ■

**Claim 2** *If  $a_B(x, \delta) > 0$ , the sign of  $\partial\Lambda_n/\partial t$  is the sign of*

$$Q_n(p) = -(2n-1)p^2 + \frac{n-2}{2}p + \frac{\delta^2}{4} - \frac{1}{16},$$

where

$$t = (1-\delta)x, \quad p = a_A(x, \delta)a_B(x, \delta) = \frac{\delta^2}{4} - (1-\delta)^2x^2.$$

Moreover, for all sufficiently large  $n$ :

- if  $0 < \delta < 1/2$ , then  $\Lambda_n(\cdot, \delta)$  has at most one local maximum;
- if  $1/2 < \delta < 1$ , then  $\Lambda_n(\cdot, \delta)$  is strictly increasing for  $x \in (0, 1/2)$ .

**Proof.** Throughout the proof, use

$$m = \frac{\delta}{2}, \quad a = m + t = a_A(x, \delta), \quad b = m - t = a_B(x, \delta).$$

Since  $s = t/m$  and  $m > 0$ , differentiating with respect to  $s$  or  $t$  gives the same sign. It is convenient to differentiate with respect to  $t$ . Since  $a = m + t$  and  $b = m - t$ , we have  $da/dt = 1$  and  $db/dt = -1$ . Using the log representation

$$\Lambda_n = \log \frac{\rho_B}{\rho_A} + \log \frac{b}{a} + (n-1) \log \frac{\frac{1}{4} - b^2}{\frac{1}{4} - a^2},$$

we obtain

$$\frac{\partial \Lambda_n}{\partial t} = - \left( \frac{1}{b} + \frac{1}{a} \right) + (n-1) \left[ \frac{2b}{\frac{1}{4} - b^2} + \frac{2a}{\frac{1}{4} - a^2} \right].$$

Equivalently,

$$\frac{\partial \Lambda_n}{\partial t} = 2(n-1) \left[ \frac{b}{\frac{1}{4} - b^2} + \frac{a}{\frac{1}{4} - a^2} \right] - \left( \frac{1}{b} + \frac{1}{a} \right).$$

We now simplify the two terms. First,

$$\begin{aligned}\frac{b}{\frac{1}{4} - b^2} + \frac{a}{\frac{1}{4} - a^2} &= \frac{b(\frac{1}{4} - a^2) + a(\frac{1}{4} - b^2)}{(\frac{1}{4} - a^2)(\frac{1}{4} - b^2)} \\ &= \frac{\frac{1}{4}(a + b) - ab(a + b)}{(\frac{1}{4} - a^2)(\frac{1}{4} - b^2)} \\ &= \frac{\delta(\frac{1}{4} - p)}{(\frac{1}{4} - a^2)(\frac{1}{4} - b^2)}.\end{aligned}$$

Second,

$$\frac{1}{b} + \frac{1}{a} = \frac{a + b}{ab} = \frac{\delta}{p}.$$

Since  $\delta > 0$ ,  $p > 0$ , and  $(\frac{1}{4} - a^2)(\frac{1}{4} - b^2) > 0$ , multiplication of  $\frac{\partial \Lambda_n}{\partial t}$  by

$$\frac{p}{\delta} \left( \frac{1}{4} - a^2 \right) \left( \frac{1}{4} - b^2 \right)$$

shows that the sign of  $\partial \Lambda_n / \partial t$  is the sign of

$$2(n-1)p \left( \frac{1}{4} - p \right) - \left( \frac{1}{4} - a^2 \right) \left( \frac{1}{4} - b^2 \right).$$

It remains to express the last product in terms of  $p$ . We have

$$\left( \frac{1}{4} - a^2 \right) \left( \frac{1}{4} - b^2 \right) = \frac{1}{16} - \frac{1}{4}(a^2 + b^2) + a^2 b^2.$$

Since

$$a^2 + b^2 = (a + b)^2 - 2ab = \delta^2 - 2p,$$

and  $a^2 b^2 = p^2$ , this becomes

$$\left( \frac{1}{4} - a^2 \right) \left( \frac{1}{4} - b^2 \right) = \frac{1}{16} - \frac{\delta^2}{4} + \frac{p}{2} + p^2.$$

Therefore the sign of  $\partial \Lambda_n / \partial t$  is the sign of

$$2(n-1)p \left( \frac{1}{4} - p \right) - \left( \frac{1}{16} - \frac{\delta^2}{4} + \frac{p}{2} + p^2 \right).$$

Expanding,

$$2(n-1)p \left( \frac{1}{4} - p \right) = \frac{n-1}{2}p - 2(n-1)p^2.$$

Thus the preceding expression equals

$$-(2n-1)p^2 + \frac{n-2}{2}p + \frac{\delta^2}{4} - \frac{1}{16} = Q_n(p),$$

as claimed.

Now suppose  $0 < \delta < 1/2$ . If  $a_B(x, \delta) > 0$ , then

$$0 < x < \frac{\delta}{2(1-\delta)}.$$

Hence

$$0 < (1-\delta)^2 x^2 < \frac{\delta^2}{4}, \quad \text{so} \quad p = \frac{\delta^2}{4} - (1-\delta)^2 x^2 \in \left(0, \frac{\delta^2}{4}\right).$$

Since  $\delta < 1/2$ , this implies  $p \in (0, 1/16)$ . Differentiating  $Q_n$  gives

$$Q'_n(p) = -2(2n-1)p + \frac{n-2}{2}.$$

Since  $p < 1/16$ ,

$$Q'_n(p) > -\frac{2n-1}{8} + \frac{n-2}{2} = \frac{2n-7}{8} > 0$$

for all sufficiently large  $n$ . Hence  $Q_n$  is increasing in  $p$ . Since

$$p = \frac{\delta^2}{4} - (1-\delta)^2 x^2$$

is decreasing in  $x$ , and hence in  $t$ , the derivative  $\partial\Lambda_n/\partial t$  changes sign at most once, from positive to negative. Therefore  $\Lambda_n(\cdot, \delta)$  is either decreasing throughout, or else increasing and then decreasing. In particular, it has at most one local maximum.

Now suppose  $1/2 \leq \delta < 1$ . For every  $x \in (0, 1/2)$ ,

$$a_B(x, \delta) = \frac{\delta}{2} - (1-\delta)x > \frac{\delta}{2} - \frac{1-\delta}{2} = \frac{2\delta-1}{2} \geq 0.$$

Thus  $a_B(x, \delta) > 0$  for all  $x \in (0, 1/2)$ . The variable  $p$  ranges from

$$p_0 = \frac{\delta^2}{4} \quad \text{as } x \downarrow 0$$

down to

$$p_1 = \frac{2\delta-1}{4} \quad \text{as } x \uparrow \frac{1}{2}.$$

The quadratic  $Q_n$  is concave in  $p$ , so its minimum on  $[p_1, p_0]$  is attained at one

of the endpoints. Substituting  $p_0 = \delta^2/4$  gives

$$\begin{aligned} Q_n(p_0) &= -(2n-1)\frac{\delta^4}{16} + \frac{n-2}{2}\frac{\delta^2}{4} + \frac{\delta^2}{4} - \frac{1}{16} \\ &= \frac{-(2n-1)\delta^4 + 2n\delta^2 - 1}{16}. \end{aligned}$$

Since  $\delta < 1$ , we see that  $Q_n(p_0) > 0$  for all sufficiently large  $n$ . Similarly, substituting  $p_1 = (2\delta - 1)/4$  gives

$$Q_n(p_1) = -(2n-1)\frac{(2\delta-1)^2}{16} + \frac{n-2}{2}\frac{2\delta-1}{4} + \frac{\delta^2}{4} - \frac{1}{16}$$

Since  $\frac{1}{2} < \delta < 1$ , we see that  $Q_n(p_1) > 0$  for  $n$  sufficiently large. Concavity implies

$$Q_n(p) > 0$$

for every  $x \in (0, 1/2)$ . Therefore  $\partial\Lambda_n/\partial t > 0$ , so  $\Lambda_n(\cdot, \delta)$  is strictly increasing for  $x \in (0, 1/2)$ . ■

**Claim 3** *For all sufficiently large  $n$ , there exists  $\delta_{1,n} \in (0, 1/2)$  such that:*

$$0 < \delta < \delta_{1,n} \implies \text{MB}_n(\cdot, \delta) \text{ is strictly decreasing for } x \in (0, 1/2),$$

and

$$\delta_{1,n} < \delta < 1/2 \implies \text{MB}_n(\cdot, \delta) \text{ is strictly decreasing, strictly increasing, then strictly decreasing.}$$

**Proof.** For  $0 < \delta < 1/2$ , define

$$L_n(\delta) = \sup_{s \in (0,1)} \Lambda_n(s, \delta),$$

and set

$$L_n(0) = \log \frac{\rho_B}{\rho_A}.$$

We make several observations in order to define  $\delta_{1,n}$ : First,  $L_n$  is weakly increasing. This is because, for each fixed  $s \in (0, 1)$ , the function  $\Lambda_n(s, \delta)$  is increasing in



$\delta$ . Indeed, only the last term in  $\Lambda_n$  depends on  $\delta$ , and

$$\begin{aligned} \frac{\partial}{\partial \delta} \log \frac{1 - \delta^2(1-s)^2}{1 - \delta^2(1+s)^2} &= -\frac{2\delta(1-s)^2}{1 - \delta^2(1-s)^2} + \frac{2\delta(1+s)^2}{1 - \delta^2(1+s)^2} \\ &= 2\delta \left[ \frac{(1+s)^2}{1 - \delta^2(1+s)^2} - \frac{(1-s)^2}{1 - \delta^2(1-s)^2} \right] > 0. \end{aligned}$$

Second,  $L_n(0) = \log \frac{\rho_B}{\rho_A} < 0$ . Third, for every fixed  $\bar{\delta} \in (0, 1/2)$ ,  $L_n(\bar{\delta}) > 0$  if  $n$  is sufficiently large. This is because, for every fixed  $s \in (0, 1)$ ,

$$\log \frac{1 - \bar{\delta}^2(1-s)^2}{1 - \bar{\delta}^2(1+s)^2} > 0,$$

so the last term in  $\Lambda_n(s, \bar{\delta})$  grows linearly in  $n$ , implying the claim.

We conclude that, for all sufficiently large  $n$ , the threshold

$$\delta_{1,n} = \inf\{\delta \in [0, 1/2) : L_n(\delta) \geq 0\}$$

is well defined and belongs to  $(0, 1/2)$ .

If  $0 < \delta < \delta_{1,n}$ , then  $\Lambda_n(s, \delta) < 0$  for every  $s \in (0, 1)$ . If  $a_B(x, \delta) > 0$ , the first claim implies  $\partial_x \text{MB}_n(x, \delta) < 0$ . If  $a_B(x, \delta) \leq 0$ , the first claim also implies  $\partial_x \text{MB}_n(x, \delta) < 0$ . Therefore  $\text{MB}_n(\cdot, \delta)$  is strictly decreasing for  $x \in (0, 1/2)$ .

Consider now  $\delta_{1,n} < \delta < 1/2$ . By the definition of the infimum, for every  $\delta > \delta_{1,n}$  there exists  $\delta_0 < \delta$  such that  $L_n(\delta_0) \geq 0$ . Since

$$\lim_{s \downarrow 0} \Lambda_n(s, \delta_0) = \log \frac{\rho_B}{\rho_A} < 0, \quad \lim_{s \uparrow 1} \Lambda_n(s, \delta_0) = -\infty,$$

the supremum defining  $L_n(\delta_0)$  is attained at some interior point  $s_0 \in (0, 1)$ . For this  $s_0$ , the map  $\delta \mapsto \Lambda_n(s_0, \delta)$  is strictly increasing. Hence

$$L_n(\delta) \geq \Lambda_n(s_0, \delta) > \Lambda_n(s_0, \delta_0) = L_n(\delta_0) \geq 0.$$

Thus  $L_n(\delta) > 0$ . By the second claim,  $\Lambda_n(\cdot, \delta)$  has at most one local maximum. Since

$$\lim_{s \downarrow 0} \Lambda_n(s, \delta) = \log \frac{\rho_B}{\rho_A} < 0, \quad \lim_{s \uparrow 1} \Lambda_n(s, \delta) = -\infty,$$

and  $L_n(\delta) > 0$ ,  $\Lambda_n$  crosses zero exactly twice in  $s \in (0, 1)$ . By the first claim, the

sign of  $\partial_x \text{MB}_n(x, \delta)$  is negative, then positive, then negative if  $a_B(x, \delta) > 0$ , i.e. for

$$x \in \left(0, \frac{\delta}{2(1-\delta)}\right).$$

For

$$x \in \left[\frac{\delta}{2(1-\delta)}, \frac{1}{2}\right),$$

we have  $a_B(x, \delta) \leq 0$ , so the first claim implies  $\partial_x \text{MB}_n(x, \delta) < 0$ . We conclude that  $\text{MB}_n(\cdot, \delta)$  is strictly decreasing, then strictly increasing, and then strictly decreasing for  $x \in (0, 1/2)$ . ■

**Claim 4** *If  $1/2 < \delta < 1$ , then  $\text{MB}_n(\cdot, \delta)$  is strictly decreasing and then strictly increasing for  $x \in (0, 1/2)$ .*

**Proof.** Fix  $1/2 < \delta < 1$ . As shown above,

$$a_B(x, \delta) > 0 \quad \text{for every } x \in (0, 1/2).$$

Hence, by the first claim, the sign of  $\partial_x \text{MB}_n(x, \delta)$  is the sign of  $\Lambda_n(s, \delta)$ , where

$$s = \frac{2(1-\delta)x}{\delta}.$$

For  $x \in (0, \frac{1}{2})$ , this variable satisfies

$$s \in (0, \frac{1-\delta}{\delta}).$$

By the second claim,  $\Lambda_n(\cdot, \delta)$  is strictly increasing. Also,

$$\lim_{s \downarrow 0} \Lambda_n(s, \delta) = \log \frac{\rho_B}{\rho_A} < 0.$$

At the other endpoint, as  $s \uparrow \frac{(1-\delta)}{\delta}$ ,

$$1 - \delta^2(1+s)^2 \downarrow 0.$$

Therefore the last logarithmic term in  $\Lambda_n(s, \delta)$  diverges to  $+\infty$ . If  $\delta > 1/2$ , the second term  $\log \frac{1-s}{1+s}$  remains finite at the upper endpoint. Hence, for all sufficiently large  $n$ ,

$$\lim_{s \uparrow (1-\delta)/\delta} \Lambda_n(s, \delta) = +\infty.$$

Thus  $\Lambda_n(\cdot, \delta)$  is strictly increasing, starts below zero, and tends to  $+\infty$ . It crosses zero exactly once. Therefore  $\partial_x \text{MB}_n(x, \delta)$  is negative before this crossing and positive after it. Hence  $\text{MB}_n(\cdot, \delta)$  is strictly decreasing and then strictly increasing for  $x \in (0, 1/2)$ . ■

Combining the preliminary case  $\delta = 0$  with Claims 3 and 4 proves the proposition.

## D Mathematical Preliminaries for the Proof of Theorem 1

**Central limit approximation.** Consider a sequence of strategies  $(\alpha_n)_{n \in \mathbb{N}}$ , and write

$$\pi_n(z) = \pi(z; \alpha_n), \quad N_n = 2n + 1.$$

For the sequences considered below,  $0 < \pi_n(z) < 1$ . Define

$$s_n(z; \alpha_n) = \left( \frac{N_n}{\pi_n(z)(1 - \pi_n(z))} \right)^{-1/2}, \quad \Delta_n(z; \alpha_n) = \frac{\pi_n(z) - \frac{n}{N_n}}{s_n(z; \alpha_n)}.$$

Thus  $s_n(z; \alpha_n)$  is the standard deviation of the vote share for  $A$  in state  $z$ , and  $\Delta_n(z; \alpha_n)$  is the distance between the expected vote share and the majority threshold, measured in standard deviations.

Let  $X_n(z) \sim \text{Bin}(N_n, \pi_n(z))$  denote the number of votes for  $A$  in state  $z$ , and set

$$Z_n(z) = \frac{X_n(z) - N_n \pi_n(z)}{\sqrt{N_n \pi_n(z)(1 - \pi_n(z))}}.$$

Since  $A$  is elected if and only if  $X_n(z) > n$ ,

$$\Pr(A \text{ is elected} \mid z; \alpha_n) = \Pr(Z_n(z) > -\Delta_n(z; \alpha_n)).$$

We use the following implication: whenever  $\Delta_n(z; \alpha_n)$  converges in  $\mathbb{R} \cup \{-\infty, +\infty\}$ ,

$$\lim_{n \rightarrow \infty} \Pr(A \text{ is elected} \mid z; \alpha_n) = \lim_{n \rightarrow \infty} \Phi(\Delta_n(z; \alpha_n)), \quad (9)$$

where  $\Phi$  is the standard normal distribution function, with  $\Phi(-\infty) = 0$  and  $\Phi(+\infty) = 1$ .

To see this, first suppose that

$$N_n \pi_n(z)(1 - \pi_n(z)) \rightarrow \infty.$$

Then the Berry–Esseen bound for Bernoulli triangular arrays implies

$$\sup_{t \in \mathbb{R}} |\Pr(Z_n(z) \leq t) - \Phi(t)| \rightarrow 0;$$

see, e.g., Feller (1971, Theorem XVI.5.1). Hence

$$\Pr(A \text{ is elected} \mid z; \alpha_n) = 1 - \Pr(Z_n(z) \leq -\Delta_n(z; \alpha_n)) = \Phi(\Delta_n(z; \alpha_n)) + o(1).$$

It remains to consider subsequences along which the variance does not diverge. Any such subsequence has a further subsequence along which  $N_n \pi_n(z)(1 - \pi_n(z))$  is bounded. Along this further subsequence,

$$\pi_n(z)(1 - \pi_n(z)) \rightarrow 0,$$

so, after passing to a further subsequence if necessary, either  $\pi_n(z) \rightarrow 0$  or  $\pi_n(z) \rightarrow 1$ .

1. If  $\pi_n(z) \rightarrow 0$ , then  $\Delta_n(z; \alpha_n) \rightarrow -\infty$ , and

$$\Pr(A \text{ is elected} \mid z; \alpha_n) = \Pr\left(\frac{X_n(z)}{N_n} > \frac{n}{N_n}\right) \leq \frac{N_n \pi_n(z)}{n} \rightarrow 0,$$

by Markov's inequality. If  $\pi_n(z) \rightarrow 1$ , then  $\Delta_n(z; \alpha_n) \rightarrow +\infty$ , and similarly

$$\Pr(A \text{ is not elected} \mid z; \alpha_n) = \Pr(N_n - X_n(z) \geq n + 1) \leq \frac{N_n(1 - \pi_n(z))}{n + 1} \rightarrow 0.$$

Thus every subsequential limit of the left-hand side of (9) agrees with the corresponding limit of  $\Phi(\Delta_n(z; \alpha_n))$ , proving (9).

**Stirling approximation.** We will repeatedly use the Stirling approximation of the pivotal likelihood in (1):

$$\Pr(\text{piv} \mid z; \alpha_n) = \frac{1 + o(1)}{\sqrt{\pi n}} \left( 4\pi(z; \alpha_n)(1 - \pi(z; \alpha_n)) \right)^n. \quad (10)$$

Thus, the pivotal likelihood is exponentially small unless  $\pi(z; \alpha_n)$  is within order  $n^{-1/2}$  of  $1/2$ . Based on the Stirling approximation, one obtains

$$\begin{aligned}\Pr(\text{piv} \mid z_A; \alpha_n) &= \frac{1 + o(1)}{\sqrt{\pi n}} \left( 4 \left( \frac{1}{2} + a_{A,n} \right) \left( \frac{1}{2} - a_{A,n} \right) \right)^n \\ &= \frac{1 + o(1)}{\sqrt{\pi n}} (1 - 4a_{A,n}^2)^n.\end{aligned}$$

where the second equality uses the differences of squares identity. Consequently,

$$\frac{\Pr(\text{piv} \mid z_A; \alpha_n)}{n^{-\frac{1}{2}}} \rightarrow \frac{1}{\sqrt{\pi}} e^{-y} \quad (11)$$

for

$$y = \lim_{n \rightarrow \infty} 4na_{A,n}^2 \in \mathbb{R} \cup \{\infty\}$$

since  $\lim_{n \rightarrow \infty} (1 - \frac{y}{n})^n = e^{-y}$  and since the exponential function is continuous.

**Uniform Bound for Pivotal Likelihood.** We repeatedly use a uniform bound for the pivotal likelihood. Namely,

$$\Pr(\text{piv} \mid z; \alpha) = \binom{2n}{n} \left( \pi(z; \alpha)(1 - \pi(z; \alpha)) \right)^n \leq \binom{2n}{n} 4^{-n}. \quad (12)$$

This holds because the function  $q(1 - q)$  has its unique maximum at  $q = \frac{1}{2}$ .

**A Generalized Poincaré–Miranda theorem.** We use the following immediate corollary of Ekmekci *et al.* (2025, Theorem 1).

**Lemma 2** *Let  $D = [\underline{x}, \bar{x}] \times [\underline{\delta}, \bar{\delta}]$ , and let  $F, G : D \rightarrow \mathbb{R}$  be continuous. Suppose that*

$$F(\underline{x}, \delta) < 0 < F(\bar{x}, \delta) \quad \text{for all } \delta \in [\underline{\delta}, \bar{\delta}],$$

*and that*

$$G(x, \underline{\delta}) > 0 \quad \text{whenever } F(x, \underline{\delta}) = 0, \quad G(x, \bar{\delta}) < 0 \quad \text{whenever } F(x, \bar{\delta}) = 0.$$

*Then there exists  $(x^*, \delta^*) \in \text{int } D$  such that*

$$F(x^*, \delta^*) = G(x^*, \delta^*) = 0.$$

**Proof.** Map  $[0, 1]^2$  onto  $D$  by

$$T(r, t) = (\underline{x} + t(\bar{x} - \underline{x}), \underline{\delta} + r(\bar{\delta} - \underline{\delta})).$$

Apply Ekmekci *et al.* (2025, Theorem 1) to

$$f(r, t) = \frac{F(T(r, t))}{1 + |F(T(r, t))|}, \quad g(r, t) = \frac{G(T(r, t))}{1 + |G(T(r, t))|}.$$

The normalization only restricts the range to  $[-1, 1]$  and preserves zeros. Theorem 1 in Ekmekci *et al.* (2025) gives an interior zero of  $f$  and  $g$ , hence an interior common zero of  $F$  and  $G$ . ■

## E Proof of Theorem 1

The first part of the theorem is the existence result of Martinelli (2006); see Theorem 4 therein. Now, we first prove the third part and then the second part of the theorem.

A useful observation is that the assumption  $C''(0) = 0$  implies

$$C'(y) = o(y) \quad \text{and} \quad C(y) = o(y^2) \quad \text{as } y \downarrow 0. \quad (13)$$

### E.1 Third part of Theorem 1

Consider any sequence of equilibria  $\alpha(x_n, \delta_n)$  with information acquisition. Write

$$a_{A,n} = a_A(x_n, \delta_n) = \frac{\delta_n}{2} + (1 - \delta_n)x_n, \quad a_{B,n} = a_B(x_n, \delta_n) = \frac{\delta_n}{2} - (1 - \delta_n)x_n.$$

We first claim that

$$\sqrt{n} a_{A,n} \rightarrow \infty.$$

Suppose not. Then, along a subsequence,  $a_{A,n} = O(n^{-1/2})$ . Since both terms in

$$a_{A,n} = \frac{\delta_n}{2} + (1 - \delta_n)x_n$$

are nonnegative, this implies  $\delta_n = O(n^{-1/2})$  and  $x_n = O(n^{-1/2})$ . By the Stirling approximation,  $P_A(x_n, \delta_n)$  is bounded below by a positive multiple of  $n^{-1/2}$  along

this subsequence. Hence

$$\text{MB}_n(x_n, \delta_n) = R_A(x_n, \delta_n) + R_B(x_n, \delta_n) \geq cn^{-1/2}$$

for some  $c > 0$ . But the first-order condition gives

$$C'(x_n) = \text{MB}_n(x_n, \delta_n),$$

whereas  $x_n = O(n^{-1/2})$  and (13) imply

$$C'(x_n) \leq C'(Kn^{-1/2}) = o(n^{-1/2})$$

for some  $K > 0$ , a contradiction. Thus  $\sqrt{n}a_{A,n} \rightarrow \infty$ .

Next we claim that

$$\sqrt{n}|a_{B,n}| \rightarrow \infty.$$

Suppose not. Then, along a subsequence,  $|a_{B,n}| = O(n^{-1/2})$ . By the Stirling approximation,  $P_B(x_n, \delta_n)$  is bounded below by a positive multiple of  $n^{-1/2}$ .

We now use the equilibrium indifference condition. Since

$$\text{MB}_n(x_n, \delta_n) = R_A(x_n, \delta_n) + R_B(x_n, \delta_n),$$

the indifference condition can be written as

$$(R_A(x_n, \delta_n) + R_B(x_n, \delta_n)) \left( \frac{1}{2} + x_n \right) - C(x_n) = R_A(x_n, \delta_n).$$

Dividing by  $R_A(x_n, \delta_n) + R_B(x_n, \delta_n)$  gives

$$\frac{R_A(x_n, \delta_n)}{R_A(x_n, \delta_n) + R_B(x_n, \delta_n)} = \frac{1}{2} + x_n - \frac{C(x_n)}{R_A(x_n, \delta_n) + R_B(x_n, \delta_n)}.$$

Using the first-order condition

$$R_A(x_n, \delta_n) + R_B(x_n, \delta_n) = C'(x_n),$$

we obtain

$$\frac{R_A(x_n, \delta_n)}{R_A(x_n, \delta_n) + R_B(x_n, \delta_n)} = \frac{1}{2} + x_n - \frac{C(x_n)}{C'(x_n)}.$$

By convexity and  $C(0) = 0$ , we have  $C(x_n) \leq x_n C'(x_n)$ , and hence the right-hand

side is at least  $1/2$ . Therefore

$$R_A(x_n, \delta_n) \geq R_B(x_n, \delta_n).$$

Since  $R_z = \rho_z P_z$ , the lower bound on  $P_B(x_n, \delta_n)$  implies that  $P_A(x_n, \delta_n)$  is also bounded below by a positive multiple of  $n^{-1/2}$ . By the Stirling approximation, this implies  $a_{A,n} = O(n^{-1/2})$ , contradicting the first claim. Thus  $\sqrt{n}|a_{B,n}| \rightarrow \infty$ .

By the central limit theorem (its implication (9)),  $\sqrt{n}a_{A,n} \rightarrow \infty$  implies that  $A$  is elected in state  $z_A$  with probability converging to one. In  $z_B$ , every subsequence has a further subsequence along which either  $a_{B,n} > 0$  or  $a_{B,n} < 0$ . In the first case,  $\sqrt{n}a_{B,n} \rightarrow \infty$ , so  $A$  is elected in  $z_B$  with probability converging to one. In the second case,  $\sqrt{n}a_{B,n} \rightarrow -\infty$ , so  $B$  is elected in  $z_B$  with probability converging to one. Therefore, if the original sequence has convergent outcome probabilities, only one of these two limiting outcome types can occur: either the sequence is asymptotically efficient, or  $A$  is elected with probability converging to one in both states.

## E.2 Second part of Theorem 1

We prove the second part by constructing a sequence of equilibria  $\alpha_n$  with information acquisition and asymptotically inefficient outcomes. Throughout, we use the notation  $\hat{x}_n$  for the unique precision that maximizes the expected utility across all strategies  $(x, A, B)$ , given a candidate strategy  $\alpha_n$ . Equivalently,  $\hat{x}_n$  solves the first-order condition

$$\text{MB}_n(x_n, \delta_n) = C'(\hat{x}_n).$$

The proof proceeds in three steps. First, we define a domain  $D_n$  of candidate strategies and relate the two equilibrium conditions (2) and (3) in Lemma 1 to roots of two functions  $F_n$  and  $G_n$ . Second, we establish the boundary conditions of the generalized Poincaré–Miranda theorem (Lemma 2) applied to  $F_n$  and  $G_n$  on  $D_n$ . Third, we use the theorem to construct an equilibrium sequence in which  $A$  is elected with probability converging to one in both states.

**Step 1: The Domain of Candidate Strategies and the Equilibrium Conditions.** We consider candidate equilibrium strategies  $\alpha_n$ , given by pairs  $(x_n, \delta_n) \in$



$D_n$  for the domain

$$D_n = \{(x, \delta) : x \in [0, Mn^{-\frac{1}{2}}], \delta \in [3Mn^{-\frac{1}{2}}, \bar{\delta}_n]\} \quad (14)$$

for some  $M > 0$  and  $\bar{\delta}_n$  close to 1 that will be defined in the course of the proof.

For  $(x, \delta) \in D_n$ , define

$$F_n(x, \delta) = \max_{\tilde{x} \in [0, 1/2]} \left\{ \text{MB}_n(x, \delta) \left( \frac{1}{2} + \tilde{x} \right) - C(\tilde{x}) - R_A(x, \delta) \right\},$$

and

$$G_n(x, \delta) = \text{MB}_n(x, \delta) - C'(x).$$

At any interior root  $(x, \delta)$  with  $F_n(x, \delta) = G_n(x, \delta) = 0$ , the equation  $G_n(x, \delta) = 0$  is exactly the first-order condition (3). Since the maximand in the definition of  $F_n$  is strictly concave in  $\tilde{x}$ , the first-order condition (3) implies that  $\tilde{x} = x$  is its unique maximizer. Hence  $F_n(x, \delta) = 0$  becomes  $\text{MB}_n(x, \delta) \left( \frac{1}{2} + x \right) - C(x) - R_A(x, \delta) = 0$ , which is precisely the indifference condition (2) between  $(x, A, B)$  and  $(0, A, A)$ . Thus the two equations  $F_n = 0$  and  $G_n = 0$  are exactly the two equilibrium conditions in Lemma 1.

## Step 2: The boundary conditions of the generalized Poincaré–Miranda theorem.

**Boundary  $x = 0$ .** Let  $x = 0$ . Then  $\pi(z_A; \alpha(0, \delta)) = \pi(z_B; \alpha(0, \delta))$  and  $P_A = P_B =: P_n(\delta)$ . Since

$$P_n(\delta) \leq \binom{2n}{n} 4^{-n} = O(n^{-1/2}),$$

the maximizer in the definition of  $F_n(0, \delta)$  converges uniformly to 0 for all  $\delta$ . Hence, uniformly,

$$F_n(0, \delta) \leq P_n(\delta) \left[ -\frac{\rho_A - \rho_B}{2} + o(1) \right] < 0$$

for all sufficiently large  $n$ .

Intuitively, since all voters strictly prefer  $A$  given the prior belief, for  $n$  sufficiently large,  $\hat{x}_n$  is sufficiently small so that after any signal  $s \in \{s_A, s_B\}$ , the voter strictly prefers  $A$  over  $B$ . Thus,  $(\hat{x}_n, A, A)$  yields strictly more utility than  $(\hat{x}_n, A, B)$ . Further,  $(0, A, A)$  yields strictly more utility than  $(\hat{x}_n, A, A)$  since the

voter does not acquire costly information. Thus, all voters strictly prefer  $(0, A, A)$  to  $(\hat{x}_n, A, B)$ , as captured by the inequality  $F_n(0, \delta) < 0$ .

**Boundary**  $x = Mn^{-\frac{1}{2}}$ . Choose  $M > 0$  so large that

$$\frac{\rho_A}{\rho_B} e^{-8M^2} < 1. \quad (15)$$

We show that this implies  $F_n(Mn^{-1/2}, \delta) > 0$  for all  $\delta \geq 3Mn^{-1/2}$  and all sufficiently large  $n$ . Let  $x = Mn^{-1/2}$ . Following the same algebra as for the Stirling approximation and since the common binomial factor cancels out,

$$\frac{P_A(x, \delta)}{P_B(x, \delta)} = \left( \frac{1 - 4a_A(x, \delta)^2}{1 - 4a_B(x, \delta)^2} \right)^n.$$

Since

$$a_A(x, \delta)^2 - a_B(x, \delta)^2 = 2\delta(1 - \delta)x, \quad (16)$$

we can rewrite this as

$$\frac{P_A(x, \delta)}{P_B(x, \delta)} = \left( 1 - \frac{a_A(x, \delta)^2 - a_B(x, \delta)^2}{\frac{1}{4} - a_B(x, \delta)^2} \right)^n = \left( 1 - \frac{2\delta(1 - \delta)x}{\frac{1}{4} - a_B(x, \delta)^2} \right)^n. \quad (17)$$

Moreover,

$$\frac{1}{4} - a_B(x, \delta)^2 = \left( \frac{1}{2} - a_B(x, \delta) \right) \left( \frac{1}{2} + a_B(x, \delta) \right).$$

Using (8), we rewrite

$$\frac{1}{2} - a_B(x, \delta) = (1 - \delta) \left( \frac{1}{2} + x \right).$$

Since  $\frac{1}{2} + a_B(x, \delta) \leq 1$ , we obtain the bound

$$\frac{1}{4} - a_B(x, \delta)^2 \leq (1 - \delta) \left( \frac{1}{2} + x \right).$$

Thus

$$\frac{P_A(x, \delta)}{P_B(x, \delta)} \leq \left( 1 - \frac{2\delta x}{\frac{1}{2} + x} \right)^n. \quad (18)$$

For  $x = Mn^{-1/2}$  and  $\delta \geq 3Mn^{-1/2}$ , we have  $\frac{2\delta x}{\frac{1}{2}+x} \geq \frac{6M^2/n}{\frac{1}{2}+Mn^{-1/2}}$ . Since, for all  $n$  sufficiently large,

$$\frac{1}{2} + Mn^{-1/2} \leq \frac{3}{4},$$

it holds

$$\frac{2\delta x}{\frac{1}{2}+x} \geq \frac{8M^2}{n},$$

so that the term on the right side of (18) is at most  $(1-8M^2/n)^n$ . Using  $\lim_{n \rightarrow \infty} (1 - \frac{y}{n})^n = e^{-y}$ ,

$$\frac{P_A(x, \delta)}{P_B(x, \delta)} \leq e^{-8M^2}.$$

By (15), this implies

$$R_A(x, \delta) < R_B(x, \delta).$$

Evaluating the maximand in the definition of  $F_n$  at  $\tilde{x} = 0$  therefore gives

$$F_n(x, \delta) \geq \frac{R_A(x, \delta) + R_B(x, \delta)}{2} - R_A(x, \delta) = \frac{R_B(x, \delta) - R_A(x, \delta)}{2} > 0.$$

Therefore  $F_n(x, \delta) > 0$  for  $x = Mn^{-1/2}$ .

**Boundary**  $\delta = 3Mn^{-\frac{1}{2}}$ . Let  $\delta = 3Mn^{-1/2}$ . It follows from (8) that, for  $x \in [0, Mn^{-1/2}]$ , the terms  $4na_A(x, \delta)^2$  and  $4na_B(x, \delta)^2$  are bounded above and below by positive constants, uniformly in  $x$ . Thus, (11) implies that  $MB_n(x, \delta)/n^{-1/2}$  is bounded below by a positive constant, uniformly in  $x$ . By (13) and convexity of  $C$ ,

$$C'(x) \leq C'(Mn^{-1/2}) = o(n^{-1/2}),$$

and

$$G_n(x, 3Mn^{-1/2}) = MB_n(x, 3Mn^{-1/2}) - C'(x) > 0$$

for all sufficiently large  $n$ .

**Boundary**  $\delta = \bar{\delta}_n$ . Choose  $\kappa > 0$  so small that

$$\frac{\rho_B}{\rho_A} e^{17\kappa} < 1. \tag{19}$$

For each sufficiently large  $n$ , choose  $\bar{\delta}_n \in (3Mn^{-1/2}, 1)$  sufficiently close to one such that  $\bar{\delta}_n \geq 1/2$  and

$$\sup_{x \in [0, Mn^{-1/2}]} \text{MB}_n(x, \bar{\delta}_n) < C'(\kappa/n). \quad (20)$$

Such a choice is possible because, for fixed  $n$ , the pivotal probabilities converge uniformly to zero on  $[0, Mn^{-1/2}]$  as  $\delta \rightarrow 1$ .

We verify the required boundary condition. Let  $x \in [0, Mn^{-1/2}]$  satisfy  $F_n(x, \bar{\delta}_n) = 0$ . We first claim that  $x > 2\kappa/n$ . Suppose instead that  $x \leq 2\kappa/n$ .

Applying steps analogous to the derivation of (17) gives

$$\frac{P_B(x, \bar{\delta}_n)}{P_A(x, \bar{\delta}_n)} = \left( 1 + \frac{a_A(x, \bar{\delta}_n)^2 - a_B(x, \bar{\delta}_n)^2}{\frac{1}{4} - a_A(x, \bar{\delta}_n)^2} \right)^n.$$

Moreover,

$$\frac{1}{4} - a_A(x, \bar{\delta}_n)^2 = \left( \frac{1}{2} - a_A(x, \bar{\delta}_n) \right) \left( \frac{1}{2} + a_A(x, \bar{\delta}_n) \right),$$

and, by (8),

$$\frac{1}{2} - a_A(x, \bar{\delta}_n) = (1 - \bar{\delta}_n) \left( \frac{1}{2} - x \right).$$

Combining these observations with (16),

$$\frac{a_A(x, \bar{\delta}_n)^2 - a_B(x, \bar{\delta}_n)^2}{\frac{1}{4} - a_A(x, \bar{\delta}_n)^2} = \frac{2\bar{\delta}_n x}{\left( \frac{1}{2} - x \right) \left( \frac{1}{2} + a_A(x, \bar{\delta}_n) \right)}.$$

If  $x \leq 2\kappa/n$ , then the numerator is at most  $4\kappa/n$ . Moreover, since  $a_A(x, \bar{\delta}_n) \geq 0$  and  $x \rightarrow 0$ , the denominator is bounded below by a number arbitrarily close to  $1/4$ . Hence, for all sufficiently large  $n$ , the last fraction is bounded above by  $17\kappa/n$ . Therefore

$$\frac{P_B(x, \bar{\delta}_n)}{P_A(x, \bar{\delta}_n)} \leq \left( 1 + \frac{17\kappa}{n} \right)^n \leq e^{17\kappa}.$$

Hence, by (19), there exists  $\eta < 1$ , independent of  $n$ , such that, for all sufficiently large  $n$ ,

$$R_B(x, \bar{\delta}_n) \leq \eta R_A(x, \bar{\delta}_n).$$

Let  $\hat{x}_n$  denote the maximizer in the definition of  $F_n(x, \bar{\delta}_n)$ . By (20), the objective in the definition of  $F_n$  has negative derivative at  $\tilde{x} = \kappa/n$ . Since this objective is strictly concave in  $\tilde{x}$ , its maximizer  $\hat{x}_n$  satisfies  $\hat{x}_n \leq \kappa/n$ . Using  $\text{MB}_n = R_A + R_B$ ,

we get

$$\begin{aligned}
F_n(x, \bar{\delta}_n) &= \text{MB}_n(x, \bar{\delta}_n) \left( \frac{1}{2} + \hat{x}_n \right) - C(\hat{x}_n) - R_A(x, \bar{\delta}_n) \\
&= -\frac{R_A(x, \bar{\delta}_n) - R_B(x, \bar{\delta}_n)}{2} + \text{MB}_n(x, \bar{\delta}_n) \hat{x}_n - C(\hat{x}_n) \\
&\leq -\frac{R_A(x, \bar{\delta}_n) - R_B(x, \bar{\delta}_n)}{2} + \text{MB}_n(x, \bar{\delta}_n) \frac{\kappa}{n}.
\end{aligned}$$

Since  $R_B(x, \bar{\delta}_n) \leq \eta R_A(x, \bar{\delta}_n)$  for some  $\eta < 1$ , the difference  $R_A(x, \bar{\delta}_n) - R_B(x, \bar{\delta}_n)$  is bounded below by a fixed positive fraction of  $\text{MB}_n(x, \bar{\delta}_n)$ . Therefore the last display is strictly negative for all sufficiently large  $n$ , contradicting  $F_n(x, \bar{\delta}_n) = 0$ . Thus  $x > 2\kappa/n$ .

At any point on the upper  $\delta$ -boundary with  $F_n(x, \bar{\delta}_n) = 0$ , we therefore have

$$\text{MB}_n(x, \bar{\delta}_n) < C'(\kappa/n) < C'(x),$$

where the second inequality follows from  $x > 2\kappa/n$  and the strict monotonicity of  $C'$ . Hence

$$G_n(x, \bar{\delta}_n) < 0.$$

**Step 3: Construction of an inefficient equilibrium sequence.** We construct an equilibrium sequence in which  $A$  is elected with probability converging to one in both states.

Applying the generalized Poincaré–Miranda theorem (Lemma 2) to  $F_n$  and  $G_n$  on the domain  $D_n$  yields, for every sufficiently large  $n$ , an interior point

$$(x_n, \delta_n) \in D_n$$

such that

$$F_n(x_n, \delta_n) = G_n(x_n, \delta_n) = 0.$$

By the discussion above, these two equations are exactly the two equilibrium conditions in Lemma 1. Hence  $\alpha(x_n, \delta_n)$  is a symmetric equilibrium with information acquisition.

Since the root lies in the interior of  $D_n$ , we have

$$0 < x_n < Mn^{-1/2} \quad \text{and} \quad \delta_n > 3Mn^{-1/2}.$$

Thus  $\delta_n > 2x_n$  for all sufficiently large  $n$ . Using (4),

$$\pi(z_B; \alpha(x_n, \delta_n)) = \frac{1}{2} + \frac{\delta_n}{2} - (1 - \delta_n)x_n > \frac{1}{2}.$$

This implies that the probability of  $A$  being elected in state  $z_B$  is strictly larger than  $\frac{1}{2}$  for all sufficiently large  $n$ . The sequence of outcome-probability pairs

$$(\Pr(A \text{ is elected} \mid z_A; \alpha(x_n, \delta_n)), \Pr(A \text{ is elected} \mid z_B; \alpha(x_n, \delta_n)))$$

is contained in the compact set  $[0, 1]^2$ . Consider any convergent subsequence. Since  $\pi(z_B; \alpha(x_n, \delta_n)) > 1/2$  along the constructed sequence, the probability that  $B$  is elected in state  $z_B$  cannot converge to one along that subsequence. Hence the subsequential limit cannot be asymptotically efficient. By the third part of Theorem 1, the only possible limit is therefore that  $A$  is elected with probability one in both states. Since every convergent subsequence has this same limit, the full constructed sequence has outcome probabilities converging to  $(1, 1)$ . In particular, the outcomes are asymptotically inefficient.

## References

- AUSTEN-SMITH, D. and BANKS, J. S. (1996). Information aggregation, rationality, and the Condorcet jury theorem. *American Political Science Review*, **90** (1), 34–45.
- BHATTACHARYA, S., DUFFY, J. and KIM, S. (2017). Voting with endogenous information acquisition: Experimental evidence. *Games and Economic Behavior*, **102**, 316–338.
- DOWNS, A. (1957). *An Economic Theory of Democracy*. New York: Harper & Row.
- DUNNING, T., GROSSMAN, G., HUMPHREYS, M., HYDE, S. D., MCINTOSH, C., NELLIS, G., ADIDA, C. L., ARIAS, E., BICALHO, C., BOAS, T. C. *et al.* (2019). Voter information campaigns and political accountability: Cumulative findings from a preregistered meta-analysis of coordinated trials. *Science Advances*, **5** (7).
- EKMEKCI, M., HEESE, C. and LAUERMANN, S. (2025). *A generalized intermediate value theorem*. Working paper.
- ELBITTAR, A., GOMBERG, A., MARTINELLI, C. and PALFREY, T. R. (2020). Ignorance and bias in collective decisions. *Journal of Economic Behavior & Organization*, **174**, 332–359.

- FEDDERSEN, T. and PESENDORFER, W. (1997). Voting behavior and information aggregation in elections with private information. *Econometrica*, **65** (5), 1029–1058.
- FELLER, W. (1971). *An Introduction to Probability Theory and Its Applications*, vol. 2. New York: John Wiley & Sons, 2nd edn.
- GERARDI, D. and YARIV, L. (2008). Information acquisition in committees. *Games and Economic Behavior*, **62** (2), 436–459.
- GERSHKOV, A. and SZENTES, B. (2009). Optimal voting schemes with costly information acquisition. *Journal of Economic Theory*, **144**, 36–68.
- GIL DE ZÚÑIGA, H., STRAUSS, N. and HUBER, B. (2020). The proliferation of the “news finds me” perception across societies. *International Journal of Communication*, **14**, 1605–1633.
- , WEEKS, B. and ARDÈVOL-ABREU, A. (2017). Effects of the news-finds-me perception in communication: Social media use implications for news seeking and learning about politics. *Journal of Computer-Mediated Communication*, **22** (3), 105–123.
- GROSSER, J. and SEEBAUER, M. (2016). The curse of uninformed voting: An experimental study. *Games and Economic Behavior*, **97**, 205–226.
- HEESE, C. (2022). *Information frictions and opposing political interests*. Working paper.
- KORIYAMA, Y. and SZENTES, B. (2009). A resurrection of the condorcet jury theorem. *Theoretical Economics*, **4** (2), 227–252.
- MARTINELLI, C. (2006). Would rational voters acquire costly information? *Journal of Economic Theory*, **129** (1), 225–251.
- (2007). Rational ignorance and voting behavior. *International Journal of Game Theory*, **35** (3), 315–335.
- MCCOMBS, M. and POINDEXTER, P. (1983). The duty to keep informed: News exposure and civic obligation. *Journal of Communication*, **33** (2), 88–96.
- MECHTENBERG, L. and TYRAN, J.-R. (2019). Voter motivation and the quality of democratic choice. *Games and Economic Behavior*, **116**, 241–259.
- MIRANDA, C. (1940). *Un’osservazione su un teorema di Brouwer*. Consiglio Nazionale delle Ricerche.
- MUKHOPADHAYA, K. (2003). Jury size and the free rider problem. *Journal of Law, Economics, and Organization*, **19** (1), 24–44.
- MYERSON, R. B. (1998). Population uncertainty and Poisson games. *International Journal of Game Theory*, **27** (3), 375–392.

- (2009). Learning from Schelling’s *Strategy of Conflict*. *Journal of Economic Literature*, **47** (4), 1109–25.
- OLIVEROS, S. (2013). *Aggregation of endogenous information in large elections*. Working paper.
- PALFREY, T. R. and ROSENTHAL, H. (1983). A strategic calculus of voting. *Public Choice*, **41** (1), 7–53.
- and — (1985). Voter participation and strategic uncertainty. *American Political Science Review*, **79** (1), 62–78.
- PALMER, R. and TOFF, B. (2020). What does it take to sustain a news habit? the role of civic duty norms and a connection to a “news community” among news avoiders in the uk and spain. *International Journal of Communication*, **14**, 1634–1653.
- PERSICO, N. (2004). Committee design with endogenous information. *The Review of Economic Studies*, **71** (1), 165–191.
- PIKETTY, T. (1999). The information-aggregation approach to political institutions. *European Economic Review*, **43** (4–6), 791–800.
- SHELLING, T. C. (1980). *The Strategy of Conflict*. Harvard University Press.
- TOFF, B. and KALOGEROPOULOS, A. (2020). All the news that’s fit to ignore: How the information environment does and does not shape news avoidance. *Public Opinion Quarterly*, **84** (S1), 366–390.
- TRIOSSI, M. (2013). Costly information acquisition. Is it better to toss a coin? *Games and Economic Behavior*, **82**, 169–191.